

Hasan Zakeri

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SUMMARY

Machine learning researcher with 8+ years of experience in autonomous vehicle systems, deep learning, and safety-critical control algorithms. Proven expertise in reinforcement learning, Bayesian neural networks, and LLM integration for intelligent systems. Skilled at applying advance scalable ML techniques in a diverse range of real-world problems. Full-stack developer with 5 years of industry experience in agile development, microservices, and responsive designs.

SKILLS AND CERTIFICATES

Machine Learning: Supervised/unsupervised learning, reinforcement learning, deep learning, statistical analysis, LLM fine-tuning, RAG, prompt engineering, transformers, graph neural networks.

Programming: Python (PyTorch, TensorFlow, Scikit-Learn, Keras, NumPy, Pandas), C++, SQL, React, Django.

Data Systems: Spark, Hadoop, ETL pipelines, large-scale data processing.

Tools & Platforms: AWS, CI/CD pipelines, MySQL, Embedded Systems.

EDA & Simulation Tools: Matlab/Simulink, CRALA, pyChrono, PyBullet, Airsim, CarSim, LTSpice, Keysight ADS, HFSS, Altium Designer, Filter Solutions.

Certificates: Designing Microsoft Azure Infrastructure Solutions (AZ-305), Implementing Analytics Solutions Using Microsoft Fabric (DP-600), Generative AI with Large Language Models, Introduction to Gen-AI, Gen-AI with Large Language Models, Build a Backend REST API with Python & Django, Python Django - The Full Stack Guide, Amateur Radio Extra Class

Soft Skills: Research, cross-functional collaboration, leadership.

EDUCATION

Illinois Institute of Technology

Ph.D. in Mechanical and Aerospace Engineering

Chicago, IL

(Expected) Jan 2026

University of Notre Dame

M.S. in Electrical Engineering

Notre Dame, IN

Advisor: Panos Antsaklis

EXPERIENCE

Illinois Institute of Technology, Graduate Research Assistant

Chicago, IL, Ongoing

- Developed a deep learning framework achieving 35% improved uncertainty quantification for autonomous vehicles.
- Built multi-agent reinforcement learning framework that improves traffic flow by 40% and reduces congestion time by 30%.

University of Notre Dame, Graduate Research Assistant

Notre Dame, IN, 2016–2019

- Designed data driven resilient controller achieving 95% attack detection accuracy using unsupervised machine learning.
- Developed polynomial approximation framework reducing computational complexity by 50% while maintaining performance.

EmNet LLC., Machine Learning Research Intern

South Bend, IN, 2016

- Created deep learning architecture improving sewer water management prediction accuracy by 40% across 4 major US cities.
- Implemented neural network control mechanism in C for EPA-SWMM, deployed in production systems across multiple cities.

Tachara Online Bookstore, Technical Project Lead

Tehran, Iran, 2010–2013

- Designed an e-commerce big data processing platform using .NET MVC 4, C# and SQL Server serving over 1M active users.
- Integrated payment gateways and inventory systems reducing transaction processing time by 60%.

Azar Negar Shargh Co., Principal Engineer

Shiraz, Iran, 2004–2009

- Architected and developed responsive, full-stack content management systems for high-profile corporate communications.
- Led cross-functional engineering teams implementing agile methodologies enabling same-day feature deployment.

MACHINE LEARNING AND CONTROL PROJECTS

Traffic Congestion using Deep RL | PyTorch, PPO, Multi-Head Attention, Actor-Critic Networks, GNN

Ongoing

- Attention-based PPO agent to optimize highway traffic flow through speed commands, achieving 25–40% congestion reduction
- Designed using multi-head attention state to handle 10–100+ vehicles with lane-change and graph-based highway topology

Distributed Filter Design Agent | Vision Transformers, YOLO, ResNet, SFT, ADS, HFSS

Winter 2024

- Trained multi-modal Vision Transformer with YOLO and ResNet on filter topologies with 92% accuracy after post-training.
- Developed end-to-end AI agent for synthesizing novel filters from desired characteristics, reducing iteration time by 85%.

RF Signal Classification Assistant | PyTorch, Transformers, Signal Processing

Spring 2024

- Developed multimodal LLM combined with spectral analysis to classify and identify 95% of unknown RF signals.
- Built conversational interface to query signal characteristics using natural language, reducing analysis time by 50%.

Autonomous Vehicles Control with RL | CARLA simulator, Stable Baselines, PyTorch, OpenCV

Spring 2023

- Trained a reinforcement learning agent using CARLA simulator images to autonomously control vehicle navigation.
- Developed a vision-based controller that learned driving in a realistic urban environment without explicit programming.

Vehicle Dynamics Model: Uncertainty Quantification | pyChrono, scikit, PyTorch, torchbnn

Winter 2022

- Validated first-principles vehicle dynamics models against Chrono::Vehicle simulations and quantified modeling uncertainty.

- Characterized modeling uncertainties using Gaussian Process Regression and Bayesian neural with 80% improvement.

Data-driven Controller Reconfiguration for Fault Mitigation

- A data-driven adaptive controller reconfiguration that detects and mitigates malfunctions using only input and output data.
- Ensures closed-loop stability and fault-tolerance in autonomous systems by a real-time interface around the existing controller

HARDWARE AND RF PROJECTS

5cm Amateur Radio Band Transverter Design | Altium Designer, Keysight ADS, Nuhertz FilterSolutions Ongoing

- Designed VCO PLL, mixer circuits, and filtering stages for a 5.7 GHz amateur radio transverter with multi-layer PCB layout.
- Achieved spurious emission suppression of -55 dBc and phase noise of -85 dBc/Hz (at 10 kHz offset).

RF Filter Synthesis Optimization | Python, scikit-rf, NumPy, SciPy, Matplotlib, MATLAB Winter 2024

- Optimization algorithms for synthesizing RF filters and component selection using manufacturer touchstone and pspice files.
- Developed an automated system that matches the desired and simulated filter response and reflection.

LCR Meter Design Using Auto-Balancing Bridge Topology Spring 2024

- Designed and implemented an impedance analyzer using auto-balancing bridge topology for RF component measurements.
- Achieved measurement accuracy of close to $\pm 0.5\%$ over frequency range of 100 Hz to 10 MHz.

SELECTED PUBLICATIONS

- **Hasan Zakeri** and Baisravan HomChaudhuri, “Graph Neural Network and Attention-Based Deep Reinforcement Learning for Heterogeneous Traffic Congestion Control with Variable Fleet Dynamics,” In preparation for The International Conference on Machine Learning (ICML 2026).
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Learning-Based Predictive Control Method for Vehicle Lateral Control with a Multi-Step Gaussian Regression Prediction,” In preparation for The Conference on Robot Learning (CoRL 2026).
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Learning-Based Predictive Control Method for Vehicle Lateral Control with a Multi-Step Gaussian Regression Prediction,” submitted, Journal of Dynamic Systems, Measurement, and Control, 2025.
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Iterative Stochastic Model Predictive Control with Multi-Step Uncertainty Learning Model: Vehicle Lane Change Case Study,” Modeling, Estimation and Control Conference, MECC 2024.
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Multi-Step Gaussian Regression Prediction in Dynamic Systems: A Case Study in Vehicle Lateral Control,” in *ASME Letters in Dynamic Systems and Control* vol. 3, no. 4, Oct. 2023.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity Measures in Cyberphysical Systems Design: An Overview of Recent Results and Applications,” in *IEEE Control Systems Magazine*, vol. 42, no. 2, pp. 118-130, April 2022.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity and Passivity Indices for Switched and Cyber-Physical Systems,” (invited) in Encyclopedia of Systems and Control, J. Baillieul and T. Samad, Eds. London: Springer London, 2020.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity, Dissipativity and Passivity Indices,” (invited) in Encyclopedia of Systems and Control, J. Baillieul and T. Samad, Eds. London: Springer London, 2020.
- **Hasan Zakeri** and Panos J. Antsaklis, Passivity and Passivity Indices of Nonlinear Systems under Operational Limitations using Approximations, International Journal of Control, July 2019.
- **Hasan Zakeri** and Panos J. Antsaklis, A Data-driven Adaptive Controller Reconfiguration for Fault Mitigation: A Passivity Approach, 27th Mediterranean Conference on Control and Automation (MED), Akko, Israel, 2019, pp. 25–30.
- More at <https://scholar.google.com/citations?user=a-Z05tMAAAAJ>

LEADERSHIP AND SERVICES

- Member of search committee for Graduate Career Director position, reviewed and interviewed 6 candidates. January 2018
University of Notre Dame
- Professional Development Chair, Graduate Student Union, University of Notre Dame 2017–2018
- Member of University Committee on Internationalization, appointed by the President, University of Notre Dame. 2017–2018
- Jury and Test Designer for Iran’s National Skills Olympiad on IT Software Solutions for Business 2013